RUTHVIK REDDY VEERANNAGARI

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OBJECTIVE

I am an aspiring researcher seeking to pursue a PhD in Computer Science with a focus on Artificial Intelligence, Machine Learning, Natural Language Processing, Computer Vision, and Extended Reality. My research interests lie in multimodal AI, knowledge representation, trustworthy intelligent systems, and immersive human-computer interaction. Through doctoral research, I aim to develop innovative computational frameworks that bridge perception, reasoning, and interaction, contributing original scientific knowledge through peer-reviewed publications and interdisciplinary collaborations. My long-term goal is to establish a research career that advances both theoretical foundations and practical applications of intelligent systems in academia and industry.

WORK EXPERIENCE

Research Intern | IIIT Naya Raipur | Jan 2026 – Present

- Developed end-to-end machine learning pipelines involving data preprocessing, feature engineering, model training, and performance evaluation.
- Conducted experimental studies using supervised learning techniques and hyperparameter optimization to improve model generalization.
- Contributed to literature review, research methodology design, reproducibility analysis, and scientific manuscript preparation.
- Collaborated with faculty researchers on AI-driven projects involving machine learning, data analytics, and intelligent systems.
- Presented research findings through technical reports, experimental evaluations, and ongoing research discussions.

PROFESSIONAL EXPERIENCE

Software Developer Intern | Cypwng, Hyderabad | Jan 2025 – Jun 2025

- Developed backend services using Django and PostgreSQL for loan management and financial workflow automation.
- Implemented OTP-based authentication, role-based access control, and secure session management mechanisms.
- Designed REST APIs enabling integration between backend services and frontend applications.
- Built modular EMI computation utilities supporting multiple loan structures and repayment schedules.

EDUCATION

Degree / Subjects	Institute / Board / University	Year	CGPA /%
Bachelor of Technology (Hons), Computer Science Engineering (Data Science)	Chhattisgarh Swami Vivekanand Technical University, Bhilai	2022–2026	7.41
Higher Secondary (12th), Physics, Chemistry, Maths	Telangana State Board of Intermediate Education, India	2020–2022	75%
10th	Telangana State Board of Secondary Education, India	2020	10.0

Projects

ArchViz-XR – AI-Powered XR Framework for Scientific Knowledge Visualization

- Developed an AI-driven semantic-to-spatial XR pipeline that converts scientific diagrams into interactive 3D augmented reality experiences.
- Integrated NLP, computer vision, and scene graph generation to automatically identify entities, relationships, and spatial layouts from research papers.
- Built an adaptive learning interface featuring AI-generated explanations, voice interaction, and immersive visualization of complex scientific concepts.
- Investigated multimodal AI and Extended Reality techniques for intelligent educational content generation.

EduChain – Blockchain-Based Academic Credential Verification

Supervisor: Mrs. Divya Shukla

- Designed a decentralized academic credential verification platform using Ethereum smart contracts for secure credential issuance and verification.
- Integrated IPFS, SHA-256 cryptographic hashing, and role-based access control to ensure tamper-resistant and verifiable academic records.
- Developed a FastAPI backend and Next.js web application enabling trustless verification for institutions, students, and employers.
- Applied blockchain technologies to improve transparency, interoperability, and security in digital credential management.

GNN-DRL & GNN-SPN – AI-Based Resource Allocation for V2X Networks

Supervisor: Prof. Rajarshi Mahapatra

- Implemented Graph Neural Network-based Deep Reinforcement Learning (GNN-DRL) and GNN-SPN models for dynamic radio resource allocation in vehicular communication networks.
- Evaluated system performance under varying traffic densities using V2I throughput, latency, interference mitigation, and reliability metrics.
- Compared graph-based reinforcement learning approaches with conventional DRL methods to improve spectrum utilization and network efficiency.
- Conducted simulation-based experiments using Python and TensorFlow for intelligent wireless resource management.

LendFundz – Loan Management and Financial Workflow Automation Platform

Supervisor: Dr. Nachiket Tapas

- Developed a fintech platform supporting end-to-end loan lifecycle management, approval workflows, EMI calculation, and repayment tracking.
- Built scalable backend services using Django and PostgreSQL with REST APIs for seamless frontend integration.
- Implemented OTP-based authentication, role-based access control, and secure session management for financial applications.
- Optimized database design and business logic to improve operational efficiency and reduce manual processing.

ParkEase – Real-Time Smart Parking Allocation System

Supervisor: Mr. Abhinaw Jagtap

- Developed a computer vision-based smart parking system for real-time parking space detection and reservation management.
- Implemented concurrency control mechanisms to prevent duplicate reservations and ensure data consistency.
- Designed automated parking allocation and live slot availability monitoring for smart-city applications.
- Applied software engineering and distributed systems principles to build a scalable parking management solution.

Mental Health Detector – Journal Sentiment Analysis System

Supervisor: Mr. Ramakant Ganjeshwar

- Developed an AI-driven sentiment analysis application to monitor emotional and behavioral patterns from personal journal entries.
- Applied Natural Language Processing techniques for text preprocessing, feature extraction, and sentiment classification.
- Built visualization dashboards to track longitudinal mood trends and behavioral changes.
- Demonstrated the application of machine learning and text analytics for personalized mental health assessment.

SOFTWARE AND RESEARCH SKILLS

Programming Languages: Python, C, C++, SQL

Frameworks & Technologies: Django, FastAPI, Next.js, REST APIs, Ethereum, Solidity, Hardhat, Ethers.js, IPFS, HTML5, CSS3, Bootstrap, JWT Authentication

Machine Learning & AI: Machine Learning, Deep Learning, Graph Neural Networks (GNN), Deep Reinforcement Learning (DRL), Natural Language Processing, Computer Vision, OpenCV, Scikit-Learn, Pandas, NumPy, Matplotlib, Feature Engineering, Hyperparameter Optimization

Wireless Communications: 5G/6G Networks, V2X Communications, Wireless Resource Allocation, Intelligent Network Management, Physical Layer Security, Internet of Things (IoT)

XR & Blockchain: Extended Reality (XR), Augmented Reality (AR), Semantic Scene Generation, 3D Scene Graph Generation, Ethereum, Smart Contracts, IPFS, SHA-256 Hashing

Databases: PostgreSQL, SQL

Tools & Platforms: Git, GitHub, Linux, Ubuntu, Windows, Jupyter Notebook, Google Colab, Postman, LaTeX

Research Domains: Artificial Intelligence, Machine Learning, Wireless Communications, Graph Learning, Natural Language Processing, Computer Vision, Extended Reality (XR), Blockchain Applications, Intelligent Systems, Trustworthy AI

Publications (Drafted / Under Review)

[1] Ruthvik Reddy Veerannagari, "EduChain: A Blockchain-Based Framework for Decentralised Academic Credential Verification," *Computer Networks*, Under Review, 2026.

[2] Ruthvik Reddy Veerannagari, "Scalable Resource Allocation Using Graph Learning in 6G for Hyper-Reliable V2X Communication," *IEEE International Conference on Advanced Networks and Telecommunications Systems (ANTS)*, Under Review, 2026.

[3] Ruthvik Reddy Veerannagari, "ArchViz-XR: AI-Powered Extended Reality Framework for Scientific Knowledge Visualization," Draft Manuscript, 2026.

ACADEMIC COURSES

B.Tech: Artificial Intelligence, Machine Learning, Pattern Recognition and Machine Learning, Natural Language Processing, Computer Vision, Intelligent Data Analysis, Big Data Analytics, Data Wrangling, Cloud Computing, Data Warehousing, Database Management Systems, Computer Networks, Cryptography and Network Security, Analysis and Design of Algorithms, Theory of Computation, Computational Complexity, Operating Systems, Computer Organization and Architecture, Software Engineering, Probability and Statistics, Data Structures, Object-Oriented Programming, Python for Data Science, R for Data Science, Data Visualization.

OTHER INTERESTS

Scientific Research, Open-Source Development, AI Hackathons, Technical Writing, Reading Research Papers, Cricket, Bike Riding.

PERSONAL DETAILS

Name	RUTHVIK REDDY VEERANNAGARI
Date of Birth	23rd August 2004
Gender	Male
Languages Known	English, Telugu, Hindi

REFERENCE

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